



EPQ DISSERTATION

Colchester Royal Grammar School

PROJECT TITLE

How will the development of technology for football impact the experience for officials, fans, and players? [5,000 words]

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Introduction:

The development of technology in football is set to significantly impact the matchday experience for officials, fans, and players in the coming years. As the sport continues to evolve, the role of technology will become increasingly important in shaping how football is played, officiated, and consumed by fans worldwide.

For these technological advancements to be beneficial, it is essential for those who are developing technology for football to consider the challenges and concerns of all stakeholders involved in the sport and how their advancements will impact upon them. From improving decision-making accuracy for referees to enhancing fan engagement and player performance, technology must be implemented with a clear focus on resolving issues that have long affected the game without worsening existing complications.

Technology is already an essential component of modern football, influencing all aspects of the sport, from the tactics and strategies used on the pitch to the way fans interact with the game at every level. Whether it's grassroots football or the spectacle of the World Cup, technological innovations are helping reshape the way the sport is experienced by players, officials, and spectators alike. In recent years, a range of technologies have been introduced to aid officials in their decision-making. The implementation of tools such as video assistant referee (VAR), goal-line technology, and instant replay systems has provided referees with access to real-time, precise data that aims to improve decision accuracy during matches. These technologies seek to reduce human error and ensure that critical moments – such as goals, penalties, and red card incidents – are reviewed thoroughly and fairly.

Ideally, these developments should lead to greater confidence in the decisions made by officials, creating a sense of trust among players and fans. However, the introduction of these technologies has currently caused more issues than they were brought in to solve. A prime example is VAR, which, while designed to reduce errors, has sparked considerable controversy. The subjectivity of certain decisions, such as offside calls or foul judgments, has resulted in confusion and frustration among fans and players alike. This ongoing debate demonstrates that the technology, while helpful, is not without its flaws, and its impact is not always as positive as intended.

Therefore, it is crucial to assess how these technologies are currently being used and evaluate their effectiveness in addressing the challenges they were designed to solve. Additionally, we must consider the perceptions of officials, fans, and players in order to understand the full extent of their impact on the matchday experience. By reflecting on the successes and shortcomings of existing technologies, we can gain valuable insights into how future technological advancements might shape the sport. This understanding will enable us to maximise the benefits of these developments while minimising any unintended consequences, ensuring that technology continues to enhance the matchday experience in a way that benefits everyone involved.

Which Technologies Have Been Integrated Into Football, And How Do They Function?

When considering technology implemented into football, people will mention one or both goal-line technology and video assistant referee (VAR). Understanding how these two technologies work, what happened during their implementation process, and why VAR has such a negative perception in football media will provide a good starting point before considering how future technology might impact the game.

Goal-Line Technology:

Goal-line technology is a system used to determine whether the ball has completely crossed the goal line and been scored in a football match. This technology is completely objective and removes the potential for human error in goal-scoring decisions.

Work began on its development after a match between England and Germany in 2010, where a shot from Frank Lampard bounced over the line but was not given as a goal by the on-field officials.

In 2011, Richard Conway¹ published an article for the BBC claiming goal-line technology could be implemented into the Premier League for the 2012/13 season. Speaking to the BBC, Alex Horne, General Secretary of the FA at the time, claimed that *“goal-line technology would be a huge boost for the game”* and that *“It’s easy to make mistakes, and we’ve all seen examples where the referee and assistant referee can’t see if a ball has crossed the line or not. We need to support them in decision-making”*.

Premier League managers welcomed the news, with Owen Coyle and Kenny Dalglish (the Bolton Wanderers and Liverpool managers) being quoted saying, *“We have the best game in the world, but the one thing I’d do to improve the game if I could, is goal-line technology”* and *“I think anything that can help get the proper decision made is going to be very well received”* respectively.

Later, in 2012, Conway² published a follow-up to his article, stating that IFAB³ would decide on goal-line technology’s fate at a meeting on July 2nd, where it would be expected to be introduced to the Laws of the Game.

In December of that year, Australia’s 3 News⁴ reported that FIFA would trial the system at the Club World Cup, with multiple systems tested across the tournament. The best system from this would then be used in the Confederations Cup in Brazil and the 2014 World Cup.

¹ Conway, R. (2011). *Premier League Could Use Goal-Line Technology in 2012-13 Season*. [online] Available at: <https://www.bbc.co.uk/sport/football/15866363> [Accessed 23 Aug. 2024].

² Conway, R. (2012). *Goal-line Technology Edges Closer*. [online] Available at: <https://www.bbc.co.uk/sport/football/17228500> [Accessed 23 Aug. 2024].

³ The International Football Association Board moderates the laws of the game. It is responsible for any law changes and must approve all technology before implementation.

⁴ 3 News (2012). *FIFA Set To Introduce Goal-Line Technology*. [online] Available at: <https://archive.ph/20130223141252/http://www.3news.co.nz/FIFA-set-to-introduce-goal-line-technology/tabid/415/articleID/279404/Default.aspx/Default.aspx> [Accessed 24 Aug. 2024].

After these trials were completed successfully, the Premier League clubs voted to adopt goal-line technology for the 2013/14 season, and the FA announced that they would have a system working at Wembley in time for the Community Shield final in August.

One implementation of goal-line technology is View 4D 2.0⁵. Vieww, the company behind the system, installed fourteen cameras and two computers in every stadium where goal-line technology is needed. Each camera works at 200 fps in full HD to record various angles of both goal mouths to provide ball tracking data, auto-zoom, and super-slow-motion videos so that goals can be automatically awarded within one second of them going in. The cameras are connected via 10 Gbps duplex connections to two servers, which forward information to a central server if and when a goal is scored, as seen in Figure 1 below.

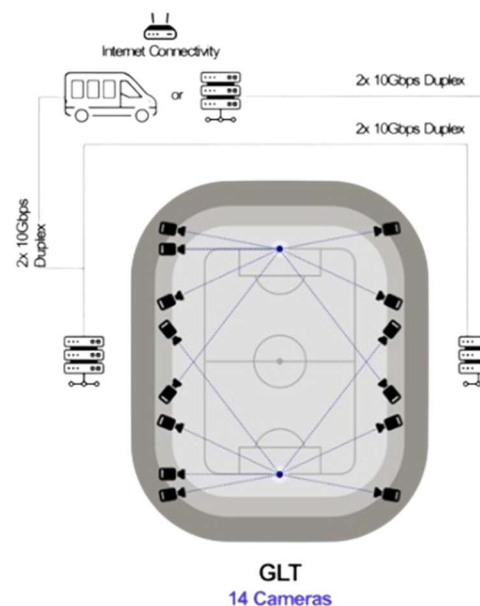


Figure 1

IFAB currently states that goal-line technology may only be applied to the goal line and can only be used to determine whether a goal has been scored. The decision from the technology must be immediate (guidelines state a one-second allowance), and it should then be communicated solely to the match officials.

⁵ vieww (2022). GLT. [online] Available at: <https://vieww.com/products/glt/>. [Accessed 23 Aug. 2024].

Video Assistant Referee:

A video assistant referee is an additional match official (or team of officials) with access to match footage and replays who can aid the onfield referee in the event of a 'clear and obvious error' with a decision made on the field or if there is a 'serious missed incident' per IFAB⁶. They designate this as any goals, penalties, direct red cards (not ones given for a second yellow card), and cases of mistaken identity.

The Laws of the Game specify that for VAR to be used, the referee must make a decision and communicate it with the VAR operator (they cannot rely on VAR to make a decision rather than make one themselves). Then, the video assistant referee operator must produce video evidence clearly showing a 'clear and obvious error' before recommending a decision to the referee. However, the final decision is always with the referee.

If the decision is subjective, the referee may be asked to go to a pitchside screen to watch the incident and to make a decision based on the footage themselves so that the operator doesn't undermine the referee's authority.

IFAB emphasises that there is no time limit for reviews as the accuracy of the final decision is always more important than the speed of the review. The only requirement is that the referee remains visible at all times to ensure transparency.

The system was first trialled in 2016 at the FIFA Club World Cup and a select group of leagues. Other leagues were only permitted to conduct 'offline' trials, where the technology would be in place, and a VAR operator would review decisions; however, they could not communicate with the on-field match officials.

One league permitted to conduct 'live' VAR tests was the Dutch Eredivisie. In an article titled *Het Jaar Van de Videoscheidsrechter* (The Year of the Video Referee) published by KNVB Media (2016)⁷, the referee for the Lichstad Derby between PSV and FC Eindhoven, Ed Janssen, stated, *"The players accepted the decision right away. That's nice to see. You make the right decision, and there is more acceptance"*. The trials continued in the KNVB Cup, where a yellow card in a match between Ajax and Willem II was upgraded to a red on the advice of VAR operator Pol van Boekel, who was quoted after the game saying, *"It is special that I was able to help the referee in this way, this shows the usefulness of the video referee"*.

⁶ The International Football Association Board (2024). *Video Assistant Referee (VAR) Protocol* | IFAB. [online] www.theifab.com. Available at: <https://www.theifab.com/laws/latest/video-assistant-referee-var-protocol/> [Accessed 18 Aug. 2024].

⁷ KNVB Media (2016). *Het Jaar Van De Videoscheidsrechter*. [online] www.knvb.nl. Available at: <https://www.knvb.nl/nieuws/themas/arbitrage-20/22846/het-jaar-van-de-videoscheidsrechter> [Accessed 19 Aug. 2024].

On March 3rd, 2018, IFAB⁸ announced that they were making a “*historic step for greater fairness in football*” by allowing FAs to permanently introduce VAR into their leagues. As reported by the BBC (2018)⁹, England FA proposed the idea to the Premier League clubs, and they agreed to implement it in time for the 2019/20 season.

A major part of the Premier League’s VAR protocols when they were originally introduced¹⁰ was the idea that VAR would only intervene if the on-field match officials made a ‘clear and obvious error’. However, what constitutes such an error was not clearly explained, leading to frustration and confusion among football fans.

In 2020, the Premier League clarified their protocols by explaining the ‘Referee Review Area’. This screen at the side of the pitch is intended for referees to view serious and subjective events such as red cards and penalties again if VAR thinks there has been an error. However, the Premier League again failed to explain to the public what warranted the referee using the screen for a review and what didn’t.

Going into the 2021/22 season, the Premier League changed its approach and very clearly explained how they had feedback from clubs and how they were adapting based on that feedback for the benefit of all involved. This involved implementing a ‘benefit of the doubt’ for attacking offsides, raising the bar for both fouls and delaying offside decisions to allow the game to flow more and to reduce the influence of VAR on games.

Going into the 2024/25 season, the Premier League introduced the idea of ‘Referee’s call’. This was inspired by cricket’s ‘Umpire’s call’ in that the original decision on the field should stand unless VAR has clear evidence showing without doubt that the on-field decision made a mistake. Otherwise, the original decision must stand.

In the vieww VAR system¹¹ each of the sixteen installed cameras acts as a server and takes one high-quality JPEG image for each frame, which is processed individually to reduce quality loss. The images are then stored on high-speed storage for client systems to request. This means that the images can be sent to the video operating room or the pitchside monitor for review.

⁸ The International Football Association Board (2018). *Historic Step for Greater Fairness in Football* | IFAB. [online] www.theifab.com. Available at: <https://www.theifab.com/news/historic-step-for-greater-fairness-in-football/> [Accessed 18 Aug. 2024].

⁹ BBC Sport (2018). *VAR Set for Premier League next Season*. [online] BBC Sport. Available at: <https://www.bbc.co.uk/sport/football/46224473> [Accessed 19 Aug. 2024].

¹⁰ Premier League (2019). *VAR: Clear and Obvious Explained*. [online] www.premierleague.com. Available at: <https://www.premierleague.com/news/1297392> [Accessed 21 Aug. 2024].

¹¹ vieww (2022). *VAR*. [online] Available at: <https://vieww.com/products/var/>. [Accessed 23rd Sept. 2024].

How Have These Technologies Influenced The Experience Of Officials, Fans And Players?

It's well known that officials, fans, and players have vastly different experiences on matchdays, so I decided to judge the impact of technology on these groups separately to best understand its true impact on the sport as a whole.

Officials:

As a referee, I sympathise with the challenge presented to professional match officials each week and the pressure from fans and pundits to make every decision correctly. As a result, I would expect officials to have a positive view of technology in football – however, this is not the whole story.

A few months ago, I attended a CPD delivered by Sam Barrott, a Premier League referee. We were given exclusive access to in-match communications between the on-field match officials. One clip was taken from a match between Bournemouth and Liverpool, where a potential offside offence occurred, which the assistant referee immediately raised their flag for.

As Barrott explained, this contradicted the Premier League's VAR protocols¹², so the referee played on. Immediately after, a potential penalty incident occurred, so trying to help the referee out from having to make an immediate decision on the penalty, the assistant raised their flag again for the original offside incident. This ended up causing chaos on the pitch as the players were visibly confused with the assistant referee's perceived indecision. Luckily for the officials, Bournemouth managed to break into a counter-attack, which resulted in a Liverpool goal-kick and time for VAR to review the incident.

Whilst this would theoretically not have happened if the officials had followed the VAR protocols, it did highlight how the often-changing protocols did create a potential for error by the match officials.

After the event, I chatted briefly with Sam, who agreed that technology was needed in football and that it mostly aids the officials. However, it also increases the pressure on the referee to make the correct decision in the first place. With Sam explaining that if VAR overturns the decision, the referee can lose some credibility with the players, which is important when trying to manage a match.

It is also important to note that having VAR does not guarantee that human error is removed from officiating. In a 2023 La Liga match between Cadiz and Elche, an Elche player scored an equaliser despite being offside in the buildup. However, none of the Cadiz players appealed at the time, and the assistant referee didn't raise their flag, so VAR chose not to go back and check for a possible offside earlier in the play. After the game, pundits and fans realised that the goal should not have stood, and Sergio Fernandez, the Cadiz manager, was quoted as saying, *"The referee said sorry,"* but that *"it wasn't much use to him by then"*.

¹² Assistant referees are expected to delay raising their flag for offsides until the end of an obvious goal scoring opportunity to allow for VAR to check the decision.

The Spanish professional refereeing body wrote to the league in response to the incident, asking them to invest in semi-automated offside technology. However, this was rejected. The referee on the day, Iglesias Villanueva, also wrote an open letter apologising to Cadiz and their fans, saying that he was “*angry, hurt, and annoyed*” because a decision of his had “*damaged Cadiz*” but that he would not accept that referees’ independence and honesty were being questioned daily. The assistant referee who missed the call (Guadalupe Porras) was reportedly in tears because they had thought it may have been offside and requested VAR to look earlier.

Overall, while the technology has been proven to increase the percentage of correct decisions for match officials, the additional pressure it brings may also negatively impact referees’ experience in football.

Fans:

When investigating technology's impact on fans, it is hard not to be drawn to social media, where you see thousands of daily posts complaining about referees and VAR. However, this was a skewed version of the truth, so I surveyed the members of CRGS to get their honest opinions on VAR, goal-line technology, and semi-automated offsides.

This is not an ideal sampling group¹³ due to its limited size and the potential influence of all the respondents having worked at or attended a grammar school, but it will suffice.

When prompted to enter a word to describe the technology in football, the top three most common answers were “*inconsistent*”, “*frustrating*”, and “*controversial*”, with forty-six negative responses and only twenty-three positive ones, as seen in Figures 2 and 3 below.



Figure 2 (Negative Response)



Figure 3 (Positive Response)

Respondents were also asked what percentage of decisions they believed goal-line technology and VAR got right. The comparison between the two is stark, with the most common answers for VAR being in the 40-80% range compared to 57% of respondents placing the accuracy of goal-line technology within the 90-100% range. This demonstrates a clear lack of trust in VAR and suggests that while technology can be successfully implemented into football, this is not the case with the implementation of VAR.

This point is backed up by respondents' quotes on VAR: “*It’s ruined the beauty and enjoyment of football from celebrating a goal like crazy to celebrating with uncertainty that some strange decision would ruin my day*” and “*It also takes away the human feel to football*”.

¹³ 72 responses, 55 ex-players, 7 managers, and 4 referees.

This contrasts with the quotes, *“This improves decision accuracy without compromising the pace or flow of the game”* and *“An enhancement to the game.”* describing goal-line technology.

Players:

Players are crucial stakeholders in football, and their opinions often shape the sport's evolution. Before the introduction of VAR, players knew that officials were making mistakes but had to accept them as part of the game. However, many were optimistic when VAR was announced to remove those errors and make officiating fairer.

In an interview in 2018, Southampton's Charlie Austin complained about a goal that had been wrongly disallowed for offside and called for the introduction of VAR to the Premier League when he demanded the league *“help the officials out – clearly they need help... we play in the best league in the world, clearly they need all the help they can get”*.

However, in 2019 (after the introduction of VAR to the Premier League), James Milner published the book *‘Ask a Footballer – My Guide to Kicking a Ball About’*. One of the topics in the book is his opinion on VAR, where he states:

“It might just be the old-school part of me, but I think there’s still too much debate around VAR. Goal-line technology is incredible. Instant decision. Black and white. But it’s very hard to use VAR when you’ve still got opinions on the decisions, and the atmosphere is being ruined. You score, there’s an explosion of noise, and then it’s VAR. You wait. Is it a goal?”

His feelings were echoed in a 2024 interview with Anthony Gordon following Newcastle's loss to Manchester United, in which he felt referee Rob Jones and VAR official Jarred Gillett denied him a penalty. After the game, Gordon said, *“I don’t mind the referee getting it wrong on the pitch, but I don’t understand the point of VAR. Either get rid of it or get better. It’s that simple, there’s too many mistakes”*.

Since the interview, Wayne Rooney has appeared on Sky Sports to say he prefers the game without VAR, saying, *“It’s taken all the enjoyment out of the game”*.

It’s clear that while players may have been optimistic about VAR before it was introduced, the vast majority now agree that they would rather go back to a time before VAR, and if an official made a mistake, they would have to accept it.

Bias Investigation:

Another point from the survey is that many fans (~45%) believe VAR to be biased. So, to test that theory, I collated a list of every time VAR has intervened in the Premier League since its introduction. For each intervention, I recorded the two teams involved, whether the home or away team was favoured, the final decision, the reason behind the intervention, and the minute in which the intervention took place.

First, I removed any teams who had played too few games for the number of VAR decisions given for or against them to be statistically significant. I defined this as one standard deviation from the mean number of games played.

$$p = \text{number of games played}, \quad n = \text{number of teams}$$

$$\bar{p} = \frac{\sum p}{n} = \frac{13(190) + 3(152) + 4(114) + 3(78) + 2(38)}{25} = 147.68$$

$$\sigma = \sqrt{\frac{\sum p^2}{n} - \bar{p}^2} = \sqrt{\frac{13(190)^2 + 3(152)^2 + 4(114)^2 + 3(78)^2 + 2(38)^2}{25} - 147.68^2} = 51.58$$

$$\text{Outliers are: } p < 96.1$$

Equation 1

Once the outliers were removed, I produced the graph shown in Figure 4 below.

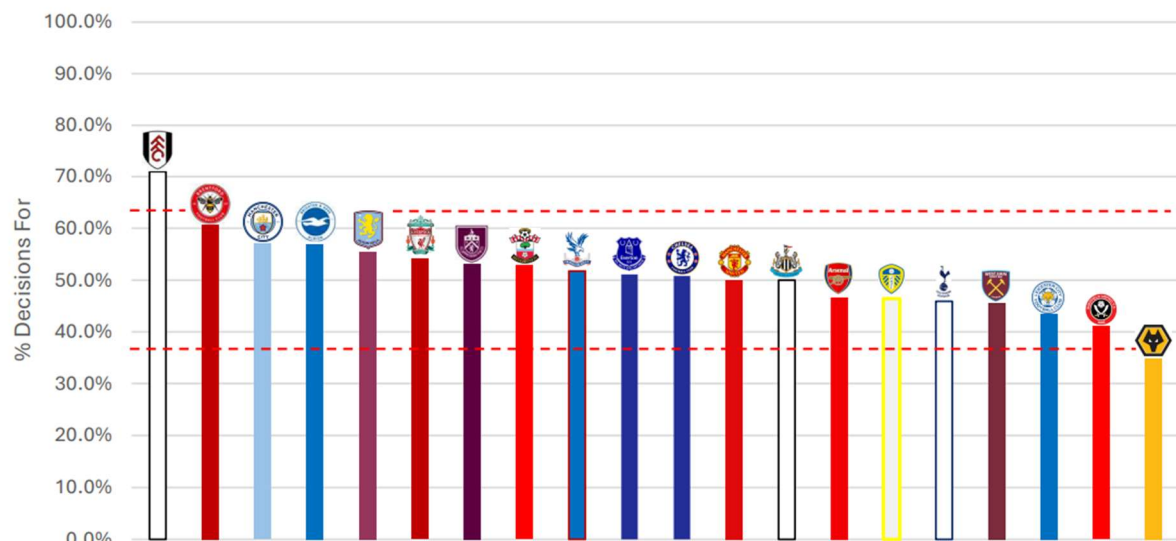


Figure 4

I then carried out the hypothesis test where the null hypothesis (H_0) was that VAR was non-biased and the chance a team gets a decision in their favour is 50%, and the alternative hypothesis (H_1) is that VAR is biased and that the proportion of decisions in a given team's favour is significantly higher or lower than 50/50. I modelled this with a binomial variable (one with two possible outcomes, as a decision will either go for or against a team).

I then set a significance level for the test as 0.002% (the percentage chance of incorrectly rejecting the null hypothesis). For this test, this is the chance of wrongly accusing PGMOL¹⁴ of being biased. I have set the value to be so low because accusing them of bias is a serious allegation, and so to do so, I would need very strong evidence.

$$X \sim B(100, p), \quad H_0: p = 0.5, \quad H_1: p \neq 0.5, \quad 0.002\% \text{ sig lvl}$$

$$P(X \leq 29) = 0.000016 = 1.6 \times 10^{-5} > 1 \times 10^{-5} \text{ REJECT}$$

$$P(X \leq 28) = 0.000006 = 0.6 \times 10^{-5} < 1 \times 10^{-5} \text{ ACCEPT}$$

$$P(X \geq 71) = 1 - P(X \leq 70) = 1 - 0.999984 = 1.6 \times 10^{-5} > 1 \times 10^{-5} \text{ REJECT}$$

$$P(X \geq 72) = 1 - P(X \leq 71) = 1 - 0.999994 = 0.6 \times 10^{-5} < 1 \times 10^{-5} \text{ ACCEPT}$$

$$\text{Critical Region: } 28 \leq X \leq 71, \quad \text{Actual Significance Level: } 0.0012\%$$

Equation 2

As no team is outside of the critical range of 28% or fewer decisions or 71% or more decisions¹⁵, there is insignificant evidence to reject the null hypothesis, and therefore, I have found VAR not to be biased.

¹⁴ Professional Game Match Officials Limited are the company responsible for providing match officials for the English Football League and Premier League.

¹⁵ See the table of data in the Appendix.

Impacts of the Technology:

It is clear that since the technology was first introduced to football, it has markedly impacted the game. For example, since the introduction of VAR, 52 goals have been given that would otherwise have been ruled out (not accounting for the net of 85 additional penalties given), and 219 goals have been ruled out after being given on-field.

VAR has also led to 44 additional red cards being handed out, with the three most common reasons for VAR to intervene being offsides (215 interventions), fouls (202 decisions), and handballs (112 interventions).

Furthermore, since the introduction of goal-line technology and VAR, there have been several notable cases of the technology breaking or being used incorrectly. Most notably for goal-line technology, in a 2020 match between Aston Villa and Sheffield United, Orjan Nyland, the Aston Villa goalkeeper, caught a free kick and fell backwards over the goal-line. However, the referee, Michael Oliver, didn't receive a signal from the goal-line technology and so didn't award a goal. The game ended 0-0, and after the game, the technology operator, Hawkeye, issued an apology and acknowledged that the ball had crossed the line but clarified that this was the first documented time where the technology failed in more than 9,000 matches.

Aston Villa finished the season one point above the relegation zone with a worse goal difference than the team in 18th, meaning this mistake contributed to keeping Aston Villa in the Premier League and relegating Bournemouth to the Championship. They wouldn't make a return to the top flight until the 2022/23 season.

In contrast, the VAR technology is fairly reliable, but this doesn't mean it isn't susceptible to human error. An infamous case of this was in the 2022/23 season, when in a match between Crystal Palace and Brighton, VAR operator John Brooks ruled out a goal by Brighton player Estupinan for an offside. However, Brooks had drawn the line on the wrong defender, having missed the position of Marc Guehi. Had the line been drawn from Guehi, Estupinan would have been onside, and the goal would have stood.

This sort of mistake only fuels fan frustrations towards VAR as the technology still doesn't prevent mistakes as was initially promised and is what has led to the calls for semi-automated offsides to be implemented into the Premier League.

In What Ways Has The Implementation Of These Technologies Led To Perceived Negative Impacts?

The difference between the perceptions of goal-line technology and VAR is stark. While goal-line technology is viewed as “*an enhancement to the game*,” VAR is described as ruining the “*beauty and enjoyment of football*.” In the survey, 90% of respondents said football is better with goal-line technology compared to 46% for VAR. So, why is this?

When comparing the implementation timelines for these technologies, goal-line technology was implemented and left as is – this has perhaps contributed to its positive perception as fans must understand the process behind the technology so that trust can be built.

In contrast, VAR has had adaptations to its protocols almost every season, and each competition that uses it has slightly differing protocols that can confuse fans and players alike. For example, when comparing the Premier League and the UEFA Champions League, both competitions that English clubs compete in, the bar for VAR to intervene for handballs differs wildly due to the different interpretations of the laws adopted by the two different bodies (PGMOL and UEFA). This difference is not publically justified and leads to the frustration of fans and players.

This is further exacerbated by the lack of transparency from IFAB. In the conditions set out for VAR use, it is specified that the communications between the referee and the VAR operator cannot be broadcast live for fans to hear. This is problematic as, unlike goal-line technology, VAR is often used for subjective decisions and, as a result, not as clear-cut for football fans to understand, so building up trust is a slower process.

PGMOL have tried to combat this heading into the 2024/25 season by setting up the X (formally known as Twitter) account ‘Premier League Match Centre’, where they post extra information about VAR interventions/times they chose not to intervene. Howard Webb is also now working with Sky Sports to release a monthly TV show, ‘Match Officials Mic’d Up,’ where they review the more controversial decisions from the month, releasing the match communications to demonstrate the process from the officials on the day before Webb explains whether he believes the decision was correct in law and inline with the spirit of football. These efforts from PGMOL provide an avenue of transparency, hopefully leading towards live-broadcast communications at some point soon.

However, as VAR protocols continue to evolve and competitions implement it in various ways, the controversy surrounding the technology is unlikely to subside. The differing thresholds for intervention and the ongoing debates about its impact on the flow of the game contribute to the ongoing scrutiny. Until a universally accepted framework is established and effectively communicated to players, officials, and fans, VAR will remain a divisive topic in football.

What Emerging Technologies Could Enhance The Experience Of Officials, Fans And Players, And How Do They Function?

Heading into the 2024/25 season, the Premier League is pushing to introduce semi-automated offsides. This technology was first trialled in the 2022 Qatar World Cup and has since been implemented in the UEFA Champions League and Serie A. It uses ball and player tracking to speed up offside decisions and remove human error by automatically calculating when the ball was kicked and where the attacker and defender's bodies were.

Genius Sports, the company contracted to implement the technology into the Premier League, uses an AI data platform which renders full 3D models of the players, the ball, and the 'offside wall'¹⁶ to enable precise offside decisions to be made quickly and accurately. Cameras around each stadium will track the movement of the ball and 10,000 data points on the bodies of all 22 players. Once a decision has been reached, a 3D animation is generated for broadcasting on stadium screens and TV.

This technology will be implemented into the league on a provisional basis, with any marginal offsides being checked using the slower, manual technology until PGMOL are convinced of the system's accuracy. Whilst this may slow the process down in the short term, it should lead to a faster, more trusted system in the future. Additionally, a limitation of the system is that it cannot determine anything subjective, such as whether an attacker is interfering with play or if a goalkeeper's vision was blocked.

In a press release from Genius Sports (2024)¹⁷, the Premier League Chief Football Officer, Tony Scholes, said that semi-automated offside "is an important move to maintain the flow of the game" and that "by using the newest and most accurate technology available, we will see the time taken for offside decisions reduced significantly alongside a consistent application of the offside lines."

This should positively impact the matchday experience for officials as it reduces what VAR officials have to check themselves when conducting a review, allowing them to focus on subjective decisions instead. This will also benefit the on-field officials as offside calls will be expedited, meaning officials will have to do less player management whilst waiting for a VAR review to conclude.

Furthermore, this change should restore some of the spontaneity and "*human aspect*" of football as goals will be quickly checked, allowing players and fans to truly enjoy their celebrations rather than anxiously awaiting the intervention of VAR. When the system was used in the Qatar World Cup¹⁸, Pierluigi Collina, FIFA's Head of Refereeing, said, "*We have come down from 70 seconds to 20 or 25. That is important.*"

¹⁶ The 'offside wall' is a graphic superimposed over the offside image which is placed across the pitch in line with the last defender's furthest back body part.

¹⁷ Genius Sports Press (2024). *Genius Sports to Supply Semi-Automated Offside Technology for the Premier League* | News | Genius Sports. [online] geniussports.com. Available at: <https://www.geniussports.com/newsroom/genius-sports-to-supply-semi-automated-offside-technology-for-the-premier-league/> [Accessed 24 Aug. 2024].

¹⁸ BBC Sport (2022). World Cup to Use semi-automated Offside Technology. BBC Sport. [online] 1 Jul. Available at: <https://www.bbc.co.uk/sport/football/62001959> [Accessed 17 Sep. 2024].

How Can The Integration Of New Technology In Football Benefit The Experience For Officials, Fans And Players?

When discussing the future of football technology, I think it is important to reiterate an extract from James Milner¹⁹: *“I think there’s still too much debate around VAR. Goal-line technology is incredible. Instant decision. Black and white.”*

This is the ethos in which we should move forward with technology development. Implemented technology should be objective and prioritise speeding up the decision-making process for officials to try and restore some of the spontaneity of football that drew so many people to the sport and that has slowly ebbed away over the past few years. Semi-automated offsides are a positive development for the sport, and similar technologies will continue to boost the experience for officials, fans, and players.

In addition, when implementing new technology, football authorities must prioritise communication with fans and players to develop trust in the technology from the beginning. This should include widely available explanations of how it will work, how it will interact with the existing Laws of the Game, and how match officials are being instructed to operate it.

This extends further to the initial trial periods carried out on developing technologies. If fans are included in the process from start to finish, it will garner respect and understanding for the process, which in turn will lead to a smoother transition when the technology is officially implemented. Transparency in testing, clear communication of its purpose, and opportunities for fan feedback will help reduce resistance and scepticism. Additionally, involving players, coaches, and referees in early trials can ensure that the technology enhances the game without disrupting its flow.

Therefore, it’s clear that the future of football technology is promising. Objective technology such as semi-automated offsides will reduce the length of reviews, improve the flow of the game, and ultimately benefit the experience of officials, fans, and players.

¹⁹ McRae, D. (2019). James Milner: ‘I’m Not a Fan of VAR... the Atmosphere Is Being Ruined’. The Guardian. [online] Available at: <https://www.theguardian.com/football/2019/oct/28/james-milner-liverpool-interview-var-premier-league> [Accessed 16 Sep. 2024].

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Appendix:

Team	Games Played	Home For	Home Against	Away For	Away Against	Total For	Total Against	% For	% Against	p90
Fulham	152	11	6	11	3	22	9	71.0%	29.0%	0.204
Nottingham Forest	76	9	5	5	2	14	7	66.7%	33.3%	0.276
Brentford	114	12	7	8	6	20	13	60.6%	39.4%	0.289
Manchester City	190	16	11	12	10	28	21	57.1%	42.9%	0.258
Brighton & Hove Albion	190	18	9	19	19	37	28	56.9%	43.1%	0.342
Aston Villa	190	12	15	19	10	31	25	55.4%	44.6%	0.295
Liverpool	190	12	13	20	14	32	27	54.2%	45.8%	0.311
Watford	76	4	4	4	3	8	7	53.3%	46.7%	0.197
Burnley	114	9	6	8	9	17	15	53.1%	46.9%	0.281
Southampton	152	5	6	14	11	19	17	52.8%	47.2%	0.237
Crystal Palace	190	10	11	19	16	29	27	51.8%	48.2%	0.295
Everton	190	11	14	12	8	23	22	51.1%	48.9%	0.237
Chelsea	190	17	14	14	16	31	30	50.8%	49.2%	0.321
Manchester United	190	16	15	16	17	32	32	50.0%	50.0%	0.337
Newcastle United	190	14	18	15	11	29	29	50.0%	50.0%	0.305
Arsenal	190	15	16	12	15	27	31	46.6%	53.4%	0.305
Leeds United	114	10	5	3	10	13	15	46.4%	53.6%	0.246
Tottenham Hotspur	190	17	18	11	15	28	33	45.9%	54.1%	0.321
West Ham United	190	10	21	20	15	30	36	45.5%	54.5%	0.347
Leicester City	152	12	12	11	18	23	30	43.4%	56.6%	0.349
Luton Town	38	2	4	1	0	3	4	42.9%	57.1%	0.184
Sheffield United	114	6	11	8	9	14	20	41.2%	58.8%	0.298
Wolverhampton Wanderers	190	10	14	6	16	16	30	34.8%	65.2%	0.242
West Bromwich Albion	38	3	4	0	5	3	9	25.0%	75.0%	0.316
Norwich City	76	4	10	1	7	5	17	22.7%	77.3%	0.289

Table 1